

The Ultimate Quant Project Pack: User Guide

45 Industry-Grade Projects for Quants, Strats, and Model Developers

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1. Why This Pack is Different (Value Proposition)

Unlike standard online courses that are often created by large teams or focus purely on academic theory, this project pack is the result of my personal experience as a Quant Analyst. I built this to mirror the actual workflow you will encounter on a trading desk.

- **Production Focus:** I don't just give you the math; I provide the **C++ implementation**, which is the industry standard for high-performance pricing libraries. Most resources stop at Python prototyping.
- **Specialized Knowledge:** I have included the "hard stuff" that actually gets you hired: **XVA (CVA/FVA), Inflation Hybrids, HJM Frameworks, and Local-Stochastic Volatility (LSV)**. These are rarely covered outside of expensive master's programs.
- **Career-Ready:** I personally wrote the "Interview Questions" and "Resume Bullets" for every single project to help you articulate your experience to hiring managers. This isn't just a code repository; it's a career accelerator built by a practitioner, for practitioners.

2. What's Inside? (The 4 Modules)

Module A: Asset Class Wise Projects (15 Projects)

- **Focus:** Specialized instruments for specific trading desks.
- **Target Audience:** Candidates interviewing for specific asset class roles (Rates, FX, Equity, etc.).
- **Key Topics:**
 - **IR:** Zero-Coupon Curve Bootstrapping, Bermudan Swaptions, Quanto Interest Rate Swaps.
 - **FX:** Forward Curve Bootstrapping, Barrier Option Pricing, Quanto Equity Options.
 - **Equity:** Dividend-Adjusted Black-Scholes, American-Asian Options, Defaultable Stock Models.
 - **Bonds:** Yield/Duration/Convexity Engine, Callable Bond Pricing, Inflation-Linked Bonds.
 - **Inflation:** CPI Simulation, Zero-Coupon Inflation Swaps, Year-on-Year Inflation Caps.
- **Goal:** Domain-specific mastery for desk-aligned roles.

Module B: Basic Projects (10 Projects)

- **Focus:** Foundations of derivatives pricing and Monte Carlo simulations.

- **Target Audience:** Interns, Students, Beginners.
- **Key Topics:** Binomial Trees, Geometric Brownian Motion (GBM), Volatility Estimation, American Put Pricing, Barrier Options, Asian Options, Greeks.
- **Goal:** Master the "Hello World" concepts of Quant Finance.

Module C: Moderate Projects (10 Projects)

- **Focus:** Intermediate numerical methods and volatility modeling.
- **Target Audience:** Junior Quants, Model Developers.
- **Key Topics:** Finite Difference Methods (Crank-Nicolson), Implied Volatility Surfaces (Dupire), Heston Model Simulation, Jump-Diffusion, Advanced Greeks.
- **Goal:** Learn how to price exotics and handle market data realities (smiles/skews).

Module D: Advanced Projects (10 Projects)

- **Focus:** Expert-level topics found on trading desks (Rates, Credit, XVA).
- **Target Audience:** Senior Quants, Desk Strats, PhDs.
- **Key Topics:** Hull-White Calibration, HJM Framework, CVA/DVA/FVA, Algorithmic Differentiation (AAD), Local-Stochastic Volatility (LSV), Credit Portfolio VaR.
- **Goal:** Solve complex, high-dimensional problems involving calibration and counterparty risk.

3. How to Use This Pack

Phase 1: The Learner (Understanding the Math)

1. Open the PDF corresponding to your level (e.g., *Asset_Class_Wise_Projects.pdf*).
2. Read the **Motivation** and **Mathematical Model** sections. Do not skip the equations; understanding the SDE/PDE is crucial.
3. **Action:** Attempt to write the Python code yourself *before* looking at the solution.

Phase 2: The Developer (Coding & Implementation)

1. **Python:** Run the provided Python scripts. Use libraries like numpy, scipy, and matplotlib to visualize the results (e.g., plot the convergence of a Monte Carlo simulation).
2. **C++:** This is the differentiator. Compile the C++ code. Focus on understanding how memory is managed and how external libraries (like Eigen or QuantLib) are integrated.
3. **Action:** Try to optimize the C++ code. Can you make the Monte Carlo engine multi-threaded using OpenMP?

Phase 3: The Candidate (Interview Prep)

1. Go to the **Interview Questions** section at the end of a project.
2. Treat these as "mock interview" questions. Can you answer them out loud without looking at the text?
3. **Resume Building:** Copy the **Enhanced Resume Bullet** for projects you have successfully understood and re-implemented. Add these to your CV to signal practical

experience.

4. Full Project List

Asset Class Wise

1. **IR:** Zero-Coupon Curve Construction & OIS Interpolation
2. **IR:** Bermudan Swaption Pricing Using LSM Monte-Carlo
3. **IR:** Quanto Interest Rate Swap Under Multi-Currency Hull-White Models
4. **FX:** FX Forward Curve Bootstrapping
5. **FX:** FX Barrier Option Pricing Using Brownian Bridge
6. **FX:** Quanto Equity Option in FX World
7. **Equity:** Dividend-Adjusted Black-Scholes Pricing Engine
8. **Equity:** American Asian Option Pricing (Penalty Method)
9. **Equity:** Equity-Credit Hybrid Defaultable Stock Model
10. **Bond:** Yield-to-Maturity, Duration, and Convexity Engine
11. **Bond:** Callable Bond Pricing via Recombining Lattice
12. **Bond:** Inflation-Linked Bond with Stochastic CPI & Credit Spread
13. **Inflation:** CPI Simulation & Zero-Coupon Inflation Swap Pricing
14. **Inflation:** Year-on-Year Inflation Cap/Floor Pricing
15. **Inflation:** Joint Inflation-Interest Rate Two-Factor Model

Basic

1. One-Period Binomial Model & Arbitrage Detection
2. GBM Simulation & Monte-Carlo Pricing
3. Historical & EWMA Volatility Estimation
4. American Put Pricing (Binomial)
5. Barrier Hitting Probability Simulation
6. Asian Option Pricing via Monte-Carlo
7. Euler-Maruyama SDE Simulation
8. Implied Volatility Root Finding (Newton's Method)
9. Exchange Option Pricing
10. Numerical Greeks (Finite Differences)

Moderate

1. Multi-Step Binomial Tree (Exotics)
2. Trinomial Tree (Adaptive)
3. Crank-Nicolson PDE Solver
4. Robust Finite-Difference Greeks (Richardson Extrapolation)
5. Merton Jump-Diffusion Model
6. Heston Stochastic Volatility (Exact Simulation)
7. Dupire Local Volatility Surface
8. Arbitrage-Free IV Surface Construction
9. American Barrier Options (Longstaff-Schwartz)
10. Heston Model Calibration

Advanced

1. Hull-White One-Factor Calibration
2. HJM Multi-Factor Monte-Carlo
3. FX Local Volatility with Quanto Adjustments
4. Reduced-Form Credit Risk (Jarrow-Turnbull)
5. Regulatory Credit Portfolio VaR (Copulas)
6. CVA/DVA & Wrong-Way Risk
7. Local-Stochastic Volatility (LSV) Calibration
8. Regime-Switching Volatility (Hamilton Filter)
9. Commodity Derivatives (Seasonal Models)
10. Advanced XVA Framework (FVA, MVA, KVA)

Disclaimer: These projects are for educational and career development purposes. They serve as a rigorous reference for standard quantitative methods used in the industry.